

Data Structures and Algorithms

Lecture 0: Course Introduction

Data Science

Beijing Normal-Hong Kong Baptist University

Course Website

- iSpace:
 - Data Structures and Algorithms (DS) (Semester 1 of 2026-2027)
 - Download lecture and lab slides
 - Check assignment information
 - Upload class exercises and assignments
 - Post your comments/questions on the forum

Grading

- Assignments
 - Written assignments (15%)
 - Programming assignments/labs (35%)
- Quiz (10%)
- Final Examination (40%)

Note:

- The final examination is very important and needs to reach at least a certain **threshold mark** in order to pass the course.
- Otherwise, the student will receive an F grade, irrespective of the marks for the other assessment components.

Reference

- Data Structures and Algorithm Analysis in C++, by Mark Allen Weiss
 - Published by Addison-Wesley, 2007
 - Later versions are all fine
 - Source codes are [available online](#)
- Data Structures and Algorithm Analysis in C++, by Clifford A. Shaffer.
 - The book itself is [available online](#).

Policies

- Submit exercises/assignments on time
 - Can submit assignments within 3 days after the deadline and get 10% off penalty each day. After 3 days, no submission will be accepted without a documented, legitimate reason.
- Academic honesty
 - Copying from others, or allowing others to copy from you, is considered academic plagiarism
 - Assignments: both get 0
 - Final: an automatic FAIL
- You are encouraged to collaborate in study groups.
 - But, you cannot copy or slightly change other students' solutions or codes. You are not allowed to copy from the Internet either. All work you submit must be your own.

Course Overview

- A fundamental computer science course
 - Essential for programming
 - Essential for advanced courses
- A challenging course, which needs
 - Mathematical and logic thinking
 - Programming skills

Course Prerequisite

- Programming
 - Need to know C
 - Visual Studio or other PC programming environment
 - Good programming skills
 - Translate pseudo-codes into codes
- Basic mathematical skills
 - Solving **recursive equations**, manipulation of symbols, etc.
- Computer architecture
 - Pointers, storage, memory access, etc.

Topics

- Algorithm Analysis
 - Mathematical Background, Big-O, Running time Calculation
- Abstract Data Types
 - Lists, Stacks and Queues
- Sorting
 - Insertion Sort, Merge Sort, Quick Sort
 - Heaps and Heap Sort
- Trees
 - Tree Traversals, Binary Trees and Binary Search Trees, AVL Trees and B+ Trees
- Graph Algorithms
 - Breath First Search

Overall Goal

- From programmer to architect
- Learn to solve problems
- Algorithms and Programming go hand in hand
- Learn to analyze your solutions